Databricks ML Associate

Questions 101–200 (Second 100)

Practice Exam Deck

How to Use This Deck

This deck contains **100 new questions** (Q101–Q200), all different from the first 100.

Each question is on its own slide with four options (A–D).

The **next slide** gives the correct answer and explains why the right option is right and why each wrong option is wrong.

Topics are spread across Feature Store, MLflow, AutoML, data processing, Spark ML, tuning, evaluation, and deployment.

Question 101

Which method retrieves a feature table object by name in the Unity Catalog Feature Store client?

A. `fe.get_table()`

B. `fe.load_table()`

C. `fe.read_table()`

D. `fe.fetch_table()`

Answer 101

Correct: A — `get_table()` returns the feature table object.

Wrong:

B is not an API.

C is not an API.

D is not an API.

Question 102

What is the purpose of a primary key in a Feature Store table?

A. It uniquely identifies each row for point-in-time lookups

B. It stores the label

C. It defines the model type

D. It sets the cluster size

Answer 102

Correct: A — primary keys identify rows for feature lookups.

Wrong:

B labels are separate.

C the model type is unrelated.

D cluster size is compute.

Question 103

Which feature table property stores the timestamp used for point-in-time joins?

A. Timestamp key

B. Primary key

C. Partition key

D. Foreign key

Answer 103

Correct: A — the timestamp key enables point-in-time lookups.

Wrong:

B is the row identifier.

C is storage layout.

D is relational, not Feature Store.

Question 104

Which method scores a batch DataFrame with a feature-aware model?

A. `fe.score_batch()`

B. `fe.predict()`

C. `fe.transform()`

D. `fe.infer()`

Answer 104

Correct: A — `score_batch()` performs feature-aware batch inference.

Wrong:

B, C, D are not the Feature Store scoring API.

Question 105

What does a FeatureLookup specify?

A. Which features to join from a feature table

B. Which model to train

C. Which cluster to use

D. Which metric to optimize

Answer 105

Correct: A — FeatureLookup defines the features to retrieve.

Wrong:

B is model training.

C is compute.

D is tuning.

Question 106

Why should feature tables be created before training sets?

A. Training sets join features from feature tables
B. Training sets create feature tables
C. They are independent
D. Training sets replace feature tables

Answer 106

Correct: A — training sets depend on feature tables.

Wrong:

B is reversed.

C is false.

D is false.

Question 107

Which MLflow component stores run metadata such as parameters and metrics?

A. The tracking server and backend store

B. The artifact store only

C. The model registry only

D. The notebook

Answer 107

Correct: A — the tracking server and backend store hold run metadata.

Wrong:

B stores artifacts.

C manages models.

D is source code.

Question 108

Which MLflow API logs a parameter?

A. `mlflow.log_param()`

B. `mlflow.log_metric()`

C. `mlflow.log_artifact()`

D. `mlflow.set_tag()`

Answer 108

Correct: A — `log_param()` logs parameters.

Wrong:

B logs metrics.

C logs artifacts.

D sets tags.

Question 109

Which MLflow API sets a tag on a run?

A. `mlflow.set_tag()`

B. `mlflow.log_tag()`

C. `mlflow.add_tag()`

D. `mlflow.tag()`

Answer 109

Correct: A — `set_tag()` sets a run tag.

Wrong:

B, C, D are not the API.

Question 110

What does `mlflow.end_run()` do?

A. Ends the active run and records its final status

B. Deletes the run

C. Starts a new run

D. Registers a model

Answer 110

Correct: A — it closes the run and records status.

Wrong:

B deletion is separate.

C is `start_run()`.

D is `register_model()`.

Question 111

Which MLflow function loads a model for inference from a registry URI?

A. `mlflow.pyfunc.load_model()`

B. `mlflow.load_run()`

C. `mlflow.get_model()`

D. `mlflow.fetch_model()`

Answer 111

Correct: A — `pyfunc.load_model()` loads a model.

Wrong:

B, C, D are not the loading API.

Question 112

What does an MLflow model signature describe?

A. The input and output schema of the model

B. The training algorithm

C. The cluster size

D. The experiment name

Answer 112

Correct: A — signatures define input/output schema.

Wrong:

B is algorithm choice.

C is compute.

D is experiment metadata.

Question 113

Which method registers a model version in Unity Catalog from a run?

A. `mlflow.register_model()`

B. `mlflow.log_model()`

C. `mlflow.save_model()`

D. `mlflow.create_model()`

Answer 113

Correct: A — `register_model()` registers a version.

Wrong:

B logs the artifact.

C saves locally.

D is not the registry API.

Question 114

What is the role of a model alias such as `@champion`?

A. It points to a specific model version for deployment
B. It deletes old versions
C. It trains the model
D. It stores metrics

Answer 114

Correct: A — aliases point to versions for serving.

Wrong:

B aliases do not delete.

C aliases do not train.

D metrics are logged separately.

Question 115

Which MLflow API searches runs in an experiment?

A. `mlflow.search_runs()`

B. `mlflow.find_runs()`

C. `mlflow.query_runs()`

D. `mlflow.list_runs()`

Answer 115

Correct: A — `search_runs()` queries runs.

Wrong:

B, C, D are not the API.

Question 116

Which AutoML API runs regression?

A. `databricks.automl.regress()`

B. `databricks.automl.classify()`

C. `databricks.automl.forecast()`

D. `databricks.automl.fit()`

Answer 116

Correct: A — `regress()` runs regression AutoML.

Wrong:

B is classification.

C is forecasting.

D is not an API.

Question 117

What does AutoML's leaderboard show?

A. Candidate models ranked by a primary metric
B. Cluster costs
C. Feature tables
D. Experiment tags

Answer 117

Correct: A — the leaderboard ranks candidate models.

Wrong:

B is billing.

C is data assets.

D is metadata.

Question 118

Which primary metric does AutoML use for classification by default?

A. F1

B. RMSE

C. MAE

D. R-squared

Answer 118

Correct: A — classification AutoML optimizes F1.

Wrong:

B, C, D are regression metrics.

Question 119

Which primary metric does AutoML use for regression by default?

A. RMSE

B. F1

C. Precision

D. Recall

Answer 119

Correct: A — regression AutoML optimizes RMSE.

Wrong:

B, C, D are classification metrics.

Question 120

Which file in an MLflow model directory defines the model's flavor and dependencies?

A. `MLmodel`

B. `model.pkl`

C. `README.md`

D. `requirements.txt`

Answer 120

Correct: A — `MLmodel` defines flavor and dependencies.

Wrong:

B is a serialized model.

C is documentation.

D is one dependency list.

Question 121

Which Spark operation selects a subset of columns?

A. `select()`

B. `filter()`

C. `groupBy()`

D. `join()`

Answer 121

Correct: A — `select()` projects columns.

Wrong:

B filters rows.

C aggregates.

D joins tables.

Question 122

Which Spark method aggregates rows by a key?

A. `groupBy()`
B. `select()`
C. `withColumn()`
D. `orderBy()`

Answer 122

Correct: A — `groupBy()` groups rows.

Wrong:

B projects columns.

C adds a column.

D sorts.

Question 123

Which method adds or replaces a column in a Spark DataFrame?

A. `withColumn()`

B. `select()`

C. `filter()`

D. `join()`

Answer 123

Correct: A — `withColumn()` adds or replaces columns.

Wrong:

B selects existing columns.

C filters rows.

D joins tables.

Question 124

Which method drops rows with nulls in any column?

A. `na.drop()`

B. `na.fill()`

C. `na.replace()`

D. `dropna("all")`

Answer 124

Correct: A — `na.drop()` drops rows with nulls.

Wrong:

B fills nulls.

C replaces values.

D is the opposite (drop only all-null rows).

Question 125

Which method fills nulls with a constant value?

A. `na.fill()`

B. `na.drop()`

C. `na.replace()`

D. `fillna()`

Answer 125

Correct: A — `na.fill()` fills nulls.

Wrong:

B drops nulls.

C replaces values.

D is pandas, not Spark.

Question 126

Why is caching a DataFrame useful in iterative ML workflows?

A. It avoids recomputing the same transformations repeatedly

B. It trains the model faster

C. It improves model accuracy

D. It reduces the feature count

Answer 126

Correct: A — caching avoids recomputation.

Wrong:

B does not affect model training speed directly.

C does not change accuracy.

D does not reduce features.

Question 127

What is the purpose of repartitioning a Spark DataFrame?

A. To change the number of partitions for parallelism
B. To drop nulls
C. To encode categoricals
D. To register a model

Answer 127

Correct: A — repartitioning controls parallelism.

Wrong:

B is null handling.

C is encoding.

D is MLflow.

Question 128

Which operation reduces the number of partitions without a shuffle when possible?

A. `coalesce()`
B. `repartition()`
C. `groupBy()`
D. `join()`

Answer 128

Correct: A — `coalesce()` reduces partitions with minimal shuffle.

Wrong:

B shuffles.

C aggregates.

D joins.

Question 129

What does `StringIndexer` output?

A. A numeric index column for each category
B. A one-hot vector
C. A scaled feature
D. A feature vector

Answer 129

Correct: A — it outputs numeric indices.

Wrong:

B is OneHotEncoder.

C is StandardScaler.

D is VectorAssembler.

Question 130

What does `OneHotEncoder` output?

A. A sparse binary vector for each category

B. A numeric index

C. A scaled feature

D. A label

Answer 130

Correct: A — one-hot outputs binary vectors.

Wrong:

B is StringIndexer.

C is scaling.

D is the target.

Question 131

Which Spark ML transformer scales features to zero mean and unit variance?

A. `StandardScaler`

B. `MinMaxScaler`

C. `MaxAbsScaler`

D. `Normalizer`

Answer 131

Correct: A — StandardScaler uses z-score scaling.

Wrong:

B scales to a range.

C scales by max absolute value.

D normalizes rows.

Question 132

Which Spark ML transformer scales features to $[0, 1]$?

A. `MinMaxScaler`

B. `StandardScaler`

C. `MaxAbsScaler`

D. `Normalizer`

Answer 132

Correct: A — MinMaxScaler scales to $[0, 1]$.

Wrong:

B is z-score.

C is max-abs.

D is row normalization.

Question 133

What is the role of a `Bucketizer` in Spark ML?

A. It bins continuous features into buckets

B. It encodes strings

C. It scales features

D. It assembles vectors

Answer 133

Correct: A — `Bucketizer` discretizes continuous features.

Wrong:

B is StringIndexer.

C is scaling.

D is VectorAssembler.

Question 134

Which Spark ML evaluator is used for binary classification?

A. `BinaryClassificationEvaluator`

B. `RegressionEvaluator`

C. `MulticlassClassificationEvaluator`

D. `ClusteringEvaluator`

Answer 134

Correct: A — it evaluates binary classifiers.

Wrong:

B is for regression.

C is for multiclass.

D is for clustering.

Question 135

Which metric can `BinaryClassificationEvaluator` compute?

A. Area under ROC
B. RMSE
C. Silhouette
D. R-squared

Answer 135

Correct: A — it computes AUC and related metrics.

Wrong:

B is regression.

C is clustering.

D is regression.

Question 136

What does the `numFolds` parameter in `CrossValidator` control?

A. The number of CV folds

B. The number of trees

C. The learning rate

D. The number of features

Answer 136

Correct: A — `numFolds` sets the number of folds.

Wrong:

B is `numTrees`.

C is `learningRate`.

D is feature count.

Question 137

What does `parallelism` in `CrossValidator` control?

A. How many models are trained concurrently
B. The number of folds
C. The number of features
D. The number of trees

Answer 137

Correct: A — it controls concurrent training.

Wrong:

B is `numFolds`.

C is feature count.

D is `numTrees`.

Question 138

Why is a `seed` passed to `CrossValidator`?

(A) To make fold splits reproducible

(B) To set the number of folds

(C) To set the learning rate

(D) To speed up training

Answer 138

Correct: A — the seed makes splits reproducible.

Wrong:

B is `numFolds`.

C is a model parameter.

D does not affect speed.

Question 139

What is the purpose of `TrainValidationSplit` in Spark ML?

A. It performs a single train-validation split for tuning
B. It performs k-fold CV
C. It trains a model without tuning
D. It deploys a model

Answer 139

Correct: A — it does a single split for tuning.

Wrong:

B is `CrossValidator`.

C is a plain fit.

D is serving.

Question 140

What does a `ParamGridBuilder` do?

A. Builds a grid of hyperparameter combinations
B. Builds a feature vector
C. Builds a pipeline
D. Builds a model

Answer 140

Correct: A — it builds a hyperparameter grid.

Wrong:

B is VectorAssembler.

C is Pipeline.

D is the estimator.

Question 141

Which Spark ML classifier supports multiclass classification natively?

A. `LogisticRegression`

B. `LinearRegression`

C. `KMeans`

D. `ALS`

Answer 141

Correct: A — Spark ML logistic regression supports multiclass.

Wrong:

B is regression.

C is clustering.

D is recommendation.

Question 142

Which Spark ML algorithm is used for collaborative filtering?

A. `ALS`

B. `KMeans`

C. `RandomForest`

D. `LogisticRegression`

Answer 142

Correct: A — ALS is used for collaborative filtering.

Wrong:

B is clustering.

C is classification/regression.

D is classification.

Question 143

Which Spark ML algorithm is a clustering method?

A. `KMeans`

B. `LinearRegression`

C. `RandomForestClassifier`

D. `ALS`

Answer 143

Correct: A — KMeans is clustering.

Wrong:

B is regression.

C is classification.

D is recommendation.

Question 144

What is the purpose of `maxIter` in Spark ML models?

A. It caps the number of optimization iterations
B. It sets the number of trees
C. It sets the learning rate
D. It sets the number of features

Answer 144

Correct: A — `maxIter` caps iterations.

Wrong:

B is `numTrees`.

C is `learningRate`.

D is feature count.

Question 145

Which Spark ML model supports feature importance?

A. `RandomForestClassifier`

B. `LinearRegression`

C. `KMeans`

D. `ALS`

Answer 145

Correct: A — tree ensembles expose feature importance.

Wrong:

B uses coefficients, not importance.

C and D do not expose importance.

Question 146

What does Hyperopt's `hp.uniform` define?

A. A uniform search space between two bounds

B. A log-uniform space

C. A choice list

D. A constant

Answer 146

Correct: A — `hp.uniform` is a uniform range.

Wrong:

B is `hp.loguniform`.

C is `hp.choice`.

D is a constant value.

Question 147

What does Hyperopt's `hp.choice` define?

A. A discrete set of options

B. A uniform range

C. A log-uniform range

D. A constant

Answer 147

Correct: A — `hp.choice` selects from options.

Wrong:

B is `hp.uniform`.

C is `hp.loguniform`.

D is a constant.

Question 148

What does `SparkTrials` require to run Hyperopt in parallel?

A. An active Spark cluster

B. A local Python environment only

C. A SQL warehouse

D. A notebook

Answer 148

Correct: A — `SparkTrials` uses a Spark cluster.

Wrong:

B is local only.

C is SQL-only.

D is not sufficient.

Question 149

Which Hyperopt search algorithm adapts based on past trials?

A. TPE

B. Random search

C. Grid search

D. Manual search

Answer 149

Correct: A — TPE adapts using previous results.

Wrong:

B, C, D do not adapt.

Question 150

Why is Bayesian optimization preferred over grid search in large search spaces?

A. It uses past results to propose promising configurations
B. It always finds the global optimum
C. It requires no objective function
D. It trains fewer models than grid search by definition

Answer 150

Correct: A — it uses past results to guide the search.

Wrong:

B no method guarantees the global optimum.

C an objective function is required.

D it can train fewer but not by definition.

Question 151

What is precision?

A. $TP/(TP + FP)$
B. $TP/(TP + FN)$
C. $TN/(TN + FP)$
D. $(TP + TN)/N$

Answer 151

Correct: A — precision is $TP/(TP + FP)$.

Wrong:

B is recall.

C is specificity.

D is accuracy.

Question 152

What is recall?

A. $TP/(TP + FN)$
B. $TP/(TP + FP)$
C. $TN/(TN + FP)$
D. $(TP + TN)/N$

Answer 152

Correct: A — recall is $TP/(TP + FN)$.

Wrong:

B is precision.

C is specificity.

D is accuracy.

Question 153

What is specificity?

A. $TN/(TN + FP)$
B. $TP/(TP + FN)$
C. $TP/(TP + FP)$
D. $(TP + TN)/N$

Answer 153

Correct: A — specificity is $TN/(TN + FP)$.

Wrong:

B is recall.

C is precision.

D is accuracy.

Question 154

What does ROC-AUC measure?

A. The probability a random positive is ranked above a random negative
B. The average precision
C. The accuracy at a fixed threshold
D. The log loss

Answer 154

Correct: A — ROC-AUC measures ranking probability.

Wrong:

B is average precision.

C is threshold-specific.

D is log loss.

Question 155

When is PR-AUC preferred over ROC-AUC?

A. Under strong class imbalance

B. When classes are balanced

C. When the target is continuous

D. When using clustering

Answer 155

Correct: A — PR-AUC is more informative under imbalance.

Wrong:

B ROC-AUC is fine when balanced.

C is regression.

D is unsupervised.

Question 156

What does a confusion matrix summarize for a binary classifier?

A. TP, FP, FN, TN counts

B. Only accuracy

C. Only precision

D. The feature importance

Answer 156

Correct: A — it holds the four confusion counts.

Wrong:

B, C are derived metrics.

D is model interpretation.

Question 157

What is the F1 score?

A. The harmonic mean of precision and recall
B. The arithmetic mean of precision and recall
C. Accuracy under imbalance
D. The area under the ROC curve

Answer 157

Correct: A — F1 is the harmonic mean of precision and recall.

Wrong:

B is not F1.

C is not F1.

D is ROC-AUC.

Question 158

Which metric is most sensitive to outliers?

A. RMSE

B. MAE

C. Median absolute error

D. R-squared

Answer 158

Correct: A — RMSE squares errors, amplifying outliers.

Wrong:

B is less sensitive.

C is robust.

D is a relative fit measure.

Question 159

What does an R^2 of 0 mean?

A. The model is no better than predicting the mean
B. The model is perfect
C. The model is overfit
D. The model has zero bias

Answer 159

Correct: A — $R^2 = 0$ matches the mean baseline.

Wrong:

B is $R^2 = 1$.

C is not indicated.

D is a bias concept.

Question 160

What does an R^2 of 1 mean?

A. The model explains all variance in the target

B. The model is no better than the mean

C. The model has zero variance

D. The model is underfit

Answer 160

Correct: A — $R^2 = 1$ is perfect fit.

Wrong:

B is $R^2 = 0$.

C is unrelated.

D is not indicated.

Question 161

What is a learning curve?

A. A plot of train and validation error versus training set size

B. A plot of accuracy versus threshold

C. A plot of precision versus recall

D. A plot of feature importance

Answer 161

Correct: A — learning curves show error vs. training size.

Wrong:

B is a threshold curve.

C is a PR curve.

D is feature importance.

Question 162

What does a persistent gap between training and validation error indicate?

A. Overfitting / high variance
B. Underfitting / high bias
C. Good generalization
D. Data leakage

Answer 162

Correct: A — a large gap indicates overfitting.

Wrong:

B underfitting shows high error on both.

C good generalization has a small gap.

D leakage is not shown by a gap alone.

Question 163

What is a validation curve?

A. A plot of train and validation score versus a hyperparameter

B. A plot of error versus training size

C. A plot of precision versus recall

D. A plot of feature importance

Answer 163

Correct: A — validation curves vary a hyperparameter.

Wrong:

B is a learning curve.

C is a PR curve.

D is feature importance.

Question 164

What is the bias-variance decomposition of expected test error?

A. $\text{Bias}^2 + \text{variance} + \text{irreducible error}$
B. $\text{Bias} + \text{variance}$
C. Training error
D. Accuracy

Answer 164

Correct: A — the decomposition includes irreducible error.

Wrong:

B omits irreducible error.

C is training error.

D is a metric.

Question 165

What is the effect of increasing regularization strength?

A. Bias increases, variance decreases

B. Bias decreases, variance increases

C. Both increase

D. Both decrease

Answer 165

Correct: A — stronger regularization raises bias and lowers variance.

Wrong:

B is less regularization.

C and D are false.

Question 166

Which method reduces variance by averaging many models?

A. Bagging
B. Boosting
C. Pruning
D. Early stopping

Answer 166

Correct: A — bagging averages models to reduce variance.

Wrong:

B boosting reduces bias.

C pruning simplifies a tree.

D stops training early.

Question 167

Which method reduces bias by sequentially correcting errors?

A. Boosting

B. Bagging

C. Pruning

D. Dropout

Answer 167

Correct: A — boosting corrects errors sequentially.

Wrong:

B is parallel.

C simplifies trees.

D is a neural net technique.

Question 168

What is the purpose of early stopping?

A. Stop training when validation performance stops improving
B. Stop training when training error reaches zero
C. Stop training after one epoch
D. Stop training when the model is too simple

Answer 168

Correct: A — early stopping uses validation performance.

Wrong:

B ignores validation.

C is arbitrary.

D is underfitting.

Question 169

What is a random forest?

A. An ensemble of decision trees trained on bootstrapped samples with random feature subsets

B. A single decision tree

C. A linear model

D. A clustering algorithm

Answer 169

Correct: A — random forests ensemble trees with bagging.

Wrong:

B is a single tree.

C is linear regression.

D is unsupervised.

Question 170

What is gradient boosting?

A. Sequential trees each fitting the previous residual
B. Parallel trees on bootstrap samples
C. A clustering method
D. A dimensionality reduction method

Answer 170

Correct: A — boosting fits residuals sequentially.

Wrong:

B is bagging.

C is clustering.

D is PCA.

Question 171

What is the main advantage of tree-based models over linear models?

A. They capture non-linear relationships and interactions
B. They always train faster
C. They need no data
D. They have no hyperparameters

Answer 171

Correct: A — trees capture non-linearity and interactions.

Wrong:

B is not always true.

C is false.

D trees have many hyperparameters.

Question 172

What is the main advantage of linear models over tree-based models?

A. Interpretability and extrapolation
B. They capture all non-linearity
C. They never overfit
D. They need no preprocessing

Answer 172

Correct: A — linear models are interpretable and extrapolate.

Wrong:

B is false; trees capture non-linearity.

C is false.

D is false; scaling is needed.

Question 173

What is k-Nearest Neighbors based on?

A. Majority/mean of the k nearest training points
B. A decision boundary learned from trees
C. A linear combination of features
D. A neural network

Answer 173

Correct: A — k-NN uses nearest neighbors.

Wrong:

B is trees.

C is linear models.

D is neural nets.

Question 174

Why must features be scaled before k-NN?

A. Distance metrics are sensitive to feature scale

B. k-NN requires categorical features

C. k-NN cannot use numeric features

D. Scaling improves accuracy only for trees

Answer 174

Correct: A — k-NN uses distances, so scaling matters.

Wrong:

B, C are false.

D scaling does not help trees.

Question 175

Which kernel is linear and used for linearly separable SVM problems?

A. Linear
B. RBF
C. Polynomial
D. Sigmoid

Answer 175

Correct: A — the linear kernel is for linear problems.

Wrong:

B, C, D are non-linear kernels.

Question 176

What does PCA produce?

A. Orthogonal components ordered by explained variance
B. Cluster labels
C. A decision boundary
D. A target variable

Answer 176

Correct: A — PCA produces orthogonal components.

Wrong:

B is clustering.

C is classification.

D is the target.

Question 177

Why is standardization important before PCA?

A. PCA maximizes variance, so unscaled features dominate
B. PCA requires categorical features
C. PCA needs no preprocessing
D. PCA only works on text

Answer 177

Correct: A — unscaled features can dominate components.

Wrong:

B, C, D are false.

Question 178

What does the silhouette score measure?

A. How similar points are to their own cluster vs. the nearest other cluster

B. Model accuracy

C. Feature importance

D. The number of clusters

Answer 178

Correct: A — silhouette measures cluster cohesion and separation.

Wrong:

B is classification.

C is interpretation.

D is a hyperparameter.

Question 179

What range does the silhouette score take?

A. $[-1, 1]$

B. $[0, 1]$

C. $[-\infty, \infty]$

D. $[0, \infty]$

Answer 179

Correct: A — silhouette ranges from -1 to 1.

Wrong:

B, C, D are incorrect ranges.

Question 180

How does DBSCAN differ from k-means?

A. DBSCAN is density-based and finds arbitrary shapes and noise
B. DBSCAN requires specifying k
C. DBSCAN is for regression
D. DBSCAN is for classification

Answer 180

Correct: A — DBSCAN is density-based and handles noise.

Wrong:

B k-means requires k.

C, D are supervised tasks.

Question 181

What is anomaly detection?

A. Identifying rare or abnormal points
B. Predicting a continuous target
C. Clustering all points
D. Reducing dimensionality

Answer 181

Correct: A — anomaly detection finds rare points.

Wrong:

B is regression.

C is clustering.

D is PCA.

Question 182

Which is an unsupervised anomaly detection method?

A. Isolation Forest

B. Logistic Regression

C. Linear Regression

D. Decision Tree Classifier

Answer 182

Correct: A — Isolation Forest is unsupervised.

Wrong:

B, C, D are supervised.

Question 183

What is the purpose of one-hot encoding?

A. Represent categorical variables as binary columns
B. Scale numeric features
C. Impute missing values
D. Reduce dimensions

Answer 183

Correct: A — one-hot represents categories as binary columns.

Wrong:

B is scaling.

C is imputation.

D is PCA.

Question 184

Why is label encoding risky for nominal categories in linear models?

A. It implies an ordinal relationship that does not exist
B. It creates too many columns
C. It removes information
D. It only works for numeric data

Answer 184

Correct: A — label encoding implies false ordering.

Wrong:

B is one-hot.

C label encoding keeps all categories.

D is false.

Question 185

What is target encoding?

A. Replacing a category with the mean target value for that category
B. One-hot encoding
C. Label encoding
D. Scaling

Answer 185

Correct: A — target encoding uses category mean targets.

Wrong:

B, C, D are different techniques.

Question 186

What is a risk of target encoding?

A. Leakage if not computed with cross-validation
B. It cannot handle categories
C. It creates sparse matrices
D. It requires no target

Answer 186

Correct: A — target encoding can leak without CV.

Wrong:

B it handles categories.

C sparse matrices are from one-hot.

D it uses the target by definition.

Question 187

What is a Pipeline in Spark ML?

A. A sequence of stages run as one estimator

B. A single model

C. A visualization tool

D. A storage format

Answer 187

Correct: A — a Pipeline chains stages.

Wrong:

B is an estimator.

C is visualization.

D is storage.

Question 188

Why should preprocessing be part of a Pipeline?

A. To fit transforms only on training data and prevent leakage
B. To speed up training
C. To reduce features
D. To register models

Answer 188

Correct: A — pipeline preprocessing prevents leakage.

Wrong:

B, C, D are not the main reason.

Question 189

Which Spark ML stage assembles multiple columns into a feature vector?

A. `VectorAssembler`

B. `StringIndexer`

C. `OneHotEncoder`

D. `StandardScaler`

Answer 189

Correct: A — VectorAssembler builds feature vectors.

Wrong:

B encodes strings.

C one-hot encodes.

D scales.

Question 190

Which stage converts a string column to numeric indices?

A. `StringIndexer`

B. `VectorAssembler`

C. `OneHotEncoder`

D. `StandardScaler`

Answer 190

Correct: A — StringIndexer converts strings to indices.

Wrong:

B assembles vectors.

C one-hot encodes.

D scales.

Question 191

What is batch inference?

A. Scoring a large dataset offline

B. Scoring one record at a time via REST

C. Training a model

D. Tuning hyperparameters

Answer 191

Correct: A — batch inference scores offline.

Wrong:

B is real-time serving.

C is training.

D is tuning.

Question 192

What is real-time inference?

A. Low-latency prediction via a REST endpoint
B. Offline scoring
C. Model training
D. Data ingestion

Answer 192

Correct: A — real-time inference serves REST predictions.

Wrong:

B is batch.

C is training.

D is ingestion.

Question 193

What is streaming inference?

A. Scoring data continuously as it arrives
B. Scoring a single batch
C. Training on a stream
D. Storing predictions

Answer 193

Correct: A — streaming inference scores continuously.

Wrong:

B is batch.

C is training.

D is storage.

Question 194

Which Databricks feature supports streaming inference?

A. Delta Live Tables
B. SQL Warehouse
C. Single Node cluster
D. Notebook widgets

Answer 194

Correct: A — Delta Live Tables supports streaming inference.

Wrong:

B is SQL only.

C is single-node.

D is UI.

Question 195

What is traffic splitting on a serving endpoint?

A. Routing a percentage of requests to different model versions
B. Training two models
C. Storing two models
D. Deleting old versions

Answer 195

Correct: A — traffic splitting routes requests.

Wrong:

B is training.

C is storage.

D is deletion.

Question 196

What is model monitoring in production?

A. Tracking performance and drift over time
B. Training the model
C. Registering the model
D. Deleting the model

Answer 196

Correct: A — monitoring tracks performance and drift.

Wrong:

B is training.

C is registration.

D is deletion.

Question 197

What is data drift?

A. A change in the distribution of input features

B. A change in the feature-target relationship

C. A model bug

D. A training error

Answer 197

Correct: A — data drift is an input distribution change.

Wrong:

B is concept drift.

C is a bug.

D is training error.

Question 198

What is concept drift?

A. A change in the feature-target relationship

B. A change in input distribution

C. A model bug

D. A data storage issue

Answer 198

Correct: A — concept drift changes the relationship.

Wrong:

B is data drift.

C is a bug.

D is storage.

Question 199

Which is a fairness concern when deploying ML?

A. Biased training data leading to discriminatory outcomes
B. Faster inference
C. Lower storage cost
D. Better dashboards

Answer 199

Correct: A — biased data causes discriminatory outcomes.

Wrong:

B, C, D are performance or cost concerns.

Question 200

Which practice best prevents data leakage in a production pipeline?

A. Fit preprocessing and the model only on training data inside a Pipeline

B. Fit preprocessing on the full dataset

C. Use the test set for feature selection

D. Scale before splitting

Answer 200

Correct: A — fitting a Pipeline on training data prevents leakage.

Wrong:

B, C, D cause leakage.